

On the Five-Point Case of Mordell's Problem

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Abstract

For five complex points $z_1, \dots, z_5$ satisfying $\sum_i |z_i|^2 = 5$, Mordell's discriminant inequality states that $\prod_{i < j} |z_i - z_j|^2 \leq 5^5$, with equality precisely for a unit regular pentagon, up to rotation and permutation. We organize its proof around a direct comparison of discriminant values. A global maximizer gives a centered stationary configuration, whose coefficient equations reduce the generic case to two quartic branches. On each branch, the absolute value of a cubic polynomial gives the original discriminant objective after accounting for the normalization scale. Exact rational interval bounds exclude both branches from the maximum, and the exceptional coefficient cases leave the regular pentagon. The argument determines the maximizing configuration without reconstructing all stationary shapes. We have completed a full Lean formalization of the five-point inequality and both directions of its equality condition, including all reductions and finite checks used in this proof.

Keywords. Mordell's inequality; discriminant; resultant; polynomial elimination; interval bounds.

1 Setup

We fix the discriminant convention, state the five-point inequality, and outline the reductions used in its proof.

1.1 Convention

For n complex points define

$$N(z) = \sum_{i=1}^n |z_i|^2, \quad \mathcal{D}_n(z) = \prod_{1 \leq i < j \leq n} |z_i - z_j|^2.$$

The constraint $N(z) = n$ fixes the total squared radius. Coincident points give $\mathcal{D}_n(z) = 0$. For $n = 5$ we write $\mathcal{D} = \mathcal{D}_5$.

1.2 Related work

The discriminant problem belongs to the study of geometric mean differences under moment constraints, including Mulholland's work on plane distributions [1]. Mordell considered the conjecture that $\mathcal{D}_n(z) \leq n^n$ when $N(z) = n$ and proved the three-point case [2]. Chalk gave counterexamples for $n \geq 6$ [3]. Kailiang Lin established the four-point case in *On Mordell's problem*, published in *Math Media* in 2011 [4].

The stationary equations also occur in the theory of relative equilibria of identical planar point vortices. Aref reviews the connection between these equilibria and polynomial roots [5],

and Dirksen and Aref study nearby relative-equilibrium configurations [6]. Faugère and Svartz give a symbolic method for polynomial systems invariant under the symmetric group and apply it to vortex equilibria [7].

Complex polynomial optimization gives another formulation of the problem. Wang and Magron include Mordell’s inequality among the examples studied with their real moment-HSOS hierarchy [8].

Wang’s exact proof, submitted on September 30, 2026 [9], establishes reflection symmetry, classifies the five collision-free equilibrium shapes, and compares their discriminants; its resultant encoding, mixed-moment identities, saturated coefficient eliminations, the dual invariant making $B = b^2$ real, and quartic branches with their coefficient graph polynomials are the algebraic inputs to the argument below.

We use these coefficient conditions to compare discriminant values directly. After normalizing a nonzero quintic coefficient, the generic case has a real invariant $B = b^2$ on one of two quartic branches. On each branch we construct a cubic polynomial V_j with $\mathcal{D}(z) = |V_j(B)|$ and prove $|V_j(B)| < 3125$ at every real root of the branch polynomial. This excludes the generic case from the maximum before any stationary shape is reconstructed. The exceptional coefficient cases then identify the regular pentagon.

1.3 Statement

From now on, $n = 5$ and $\mathcal{D} = \mathcal{D}_5$.

Theorem 1.1 (Five-point inequality). *Let $z_1, \dots, z_5 \in \mathbb{C}$ satisfy $N(z) = 5$. Then*

$$\mathcal{D}(z) \leq 3125 = 5^5. \quad (1)$$

Equality holds if and only if there are a unit complex number u , a primitive fifth root of unity ω , and a permutation σ of $\{0, 1, 2, 3, 4\}$ such that

$$z_{i+1} = u\omega^{\sigma(i)} \quad (0 \leq i \leq 4). \quad (2)$$

1.4 Proof strategy

A sharp extremal bound requires identifying the maximizing configurations. We therefore use the coefficient equations as necessary conditions and compare the objective on their branches, retaining the real invariant $B = b^2$ and the normalization scale throughout the calculation.

The argument proceeds in five stages. First, compactness and the regular pentagon give a positive global maximum; translation and scaling show that it is centered. Second, differentiation of a homogeneous polynomial envelope yields the stationary equations. Third, the resultant and mixed-moment identities give the coefficient constraints and the two generic quartic branches. Fourth, reduction modulo each quartic produces a cubic polynomial for the discriminant value, and rational interval bounds exclude every real root from attaining 3125. Fifth, direct calculations in the exceptional coefficient cases leave $X^5 + d$, whose roots form a unit regular pentagon.

The generic comparison uses the identity that makes B real, then passes directly to the discriminant. It replaces the subsequent reflection-symmetric parametrizations and the classification by one, three, or five real roots with two scalar value bounds. The resulting certificates need to cover the real zero sets of the quartics, rather than identify their physical shapes. Section 2 derives the coefficient conditions, and Section 3 constructs the value bounds and determines the equality case.

2 Reductions

This section connects the original norm constraint to the polynomial conditions used by the finite certificates. We first derive stationarity at a global maximizer and then pass to the coefficients of its root polynomial.

2.1 Stationary reduction

2.1.1 Existence, positivity, and centering

The set $S = \{z \in \mathbb{C}^5 : N(z) = 5\}$ is compact, and $\mathcal{D}$ is continuous. Hence $\mathcal{D}$ has a global maximizer on S . A unit regular pentagon belongs to S and has discriminant 3125. One way to see the latter identity is to use the roots of $X^5 - 1$: the product of its derivatives at those roots is 5^5 . Thus every global maximizer satisfies

$$\mathcal{D}(z) \geq 3125 > 0. \quad (3)$$

In particular its five coordinates are pairwise distinct.

Centering follows from a translation and scaling argument. Write $m = \frac{1}{5} \sum_i z_i$ and $w_i = z_i - m$. Then

$$\mathcal{D}(w) = \mathcal{D}(z), \quad N(w) = N(z) - 5|m|^2. \quad (4)$$

For a positive-discriminant configuration, $N(w) > 0$. If $m \neq 0$, then $N(w) < 5$, so multiplying w by $r = \sqrt{5/N(w)} > 1$ restores the original constraint. Since there are ten pairs,

$$\mathcal{D}(rw) = r^{20} \mathcal{D}(w) > \mathcal{D}(z),$$

contradicting global maximality. Thus a global maximizer has $\sum_i z_i = 0$.

2.1.2 The force equation

For a collision-free configuration define

$$F_i(z) = \sum_{j \neq i} \frac{1}{z_i - z_j}.$$

For a complex direction $h \in \mathbb{C}^5$, differentiation along the real parameter s gives

$$\left. \frac{d}{ds} \mathcal{D}(z + sh) \right|_{s=0} = 2\mathcal{D}(z) \operatorname{Re} \sum_i h_i F_i(z), \quad (5)$$

$$\left. \frac{d}{ds} N(z + sh) \right|_{s=0} = 2 \operatorname{Re} \sum_i h_i \bar{z}_i. \quad (6)$$

The degree-20 homogeneity of $\mathcal{D}$ yields the unconstrained envelope

$$\mathcal{D}(w) \leq \mathcal{D}(z) \left(\frac{N(w)}{5} \right)^{10} \quad (w \in \mathbb{C}^5)$$

at a global maximizer z . Taking the derivative of the difference of the two sides at $w = z$ shows that

$$\operatorname{Re} \sum_i h_i (F_i(z) - 2\bar{z}_i) = 0 \quad (h \in \mathbb{C}^5).$$

Directions supported on one coordinate, with values 1 and i , give

$$F_i(z) = 2\bar{z}_i \quad (1 \leq i \leq 5). \quad (7)$$

2.1.3 Conjugate scaling

Coefficient normalization requires a possibly nonreal scale. For $t \neq 0$ set

$$x_i = tz_i, \quad y_i = \frac{\bar{z}_i}{t}, \quad \kappa = |t|^{-2} > 0. \quad (8)$$

Then $y_i = \kappa \bar{x}_i$, and (7) and its conjugate imply

$$\sum_{j \neq i} \frac{1}{x_i - x_j} = 2y_i, \quad \sum_{j \neq i} \frac{1}{y_i - y_j} = 2x_i. \quad (9)$$

Both configurations are centered and collision-free, and $N(x) = 5|t|^2 = 5/\kappa$.

2.2 Resultant reduction

The next step replaces the point coordinates by four quintic coefficients. The resultant carries the stationary equations into polynomial identities, while a positive scale parameter retains the original norm constraint.

2.2.1 The centered root polynomial

Let

$$p(X) = \prod_{i=1}^5 (X - x_i) = X^5 + aX^3 + bX^2 + cX + d. \quad (10)$$

The missing X^4 term follows from centering. At a simple root,

$$\frac{p''(x_i)}{2p'(x_i)} = \sum_{j \neq i} \frac{1}{x_i - x_j} = 2y_i, \quad p''(x_i) = 4y_i p'(x_i).$$

Put

$$\delta = \prod_i p'(x_i).$$

For degree five the sign in the derivative-product formula is positive, so $\delta = \prod_{i < j} (x_i - x_j)^2 \neq 0$. This complex discriminant is distinct from the nonnegative real objective $D = \mathcal{D}(z)$.

Define the auxiliary resultant

$$H(Y, T) = \text{Res}_X(p(X), (Y + TX)p'(X) - p''(X)). \quad (11)$$

Evaluation at the roots of the monic polynomial p proves

$$H(Y, T) = \delta \prod_i (Y + Tx_i - 4y_i). \quad (12)$$

Subtracting $5Tp$ from the second input reduces its degree to four. Thus its Sylvester determinant has size 9×9 , and its expansion gives the coefficient expressions used below.

2.2.2 The zero and first coefficients in T

Define polynomial coefficient expressions r_3, r_2, r_1, r_0 and s_2, s_1, s_0 by

$$H(Y, 0) = \delta Y^5 + r_3 Y^3 + r_2 Y^2 + r_1 Y + r_0, \quad (13)$$

$$\partial_T H(Y, 0) = 20\delta Y^3 + s_2 Y^2 + s_1 Y + s_0. \quad (14)$$

These are rational polynomials in a, b, c, d . The product expression (12), combined with the dual stationary equations, gives mixed-moment identities

$$\begin{aligned} \sum_i x_i y_i &= 5, & \sum_i x_i y_i^2 &= 0, \\ 4 \sum_i x_i y_i^3 &= 7 \sum_i y_i^2, & 2 \sum_i x_i y_i^4 &= 3 \sum_i y_i^3. \end{aligned}$$

They follow by summing the force equations against powers of y_i and pairing the two orientations of each unordered pair. Differentiating (12) at $T = 0$ and applying these identities proves

$$s_2 = 0, \quad s_1 = 6r_3, \quad s_0 = 2r_2. \quad (15)$$

For example, the Y^3 coefficient is $4\delta \sum_i x_i y_i = 20\delta$, and the Y^2 coefficient vanishes because $\sum_i x_i y_i^2 = 0$. The remaining two coefficients are handled by the displayed cubic and quartic mixed moments and Newton identities for the centered polynomial of the y_i .

2.2.3 Conjugation and the original discriminant

The root polynomial of $y_i = \kappa \bar{x}_i$ has coefficients $\kappa^2 \bar{a}, \kappa^3 \bar{b}, \kappa^4 \bar{c}, \kappa^5 \bar{d}$. Taking coefficients of (12) at $T = 0$ therefore gives

$$r_3 = 16\delta \kappa^2 \bar{a}, \quad r_2 = 64\delta \kappa^3 \bar{b}, \quad (16)$$

$$r_1 = 256\delta \kappa^4 \bar{c}, \quad r_0 = 1024\delta \kappa^5 \bar{d}. \quad (17)$$

Furthermore, (8) implies

$$\kappa^{20} |\delta|^2 = D^2. \quad (18)$$

Thus the coefficients of a global maximizer satisfy $\kappa > 0$, $\delta \neq 0$, (15), and (16)–(18).

3 Coefficient branches

The coefficient conditions reduce the proof to finitely many branches. We describe their exact certificates, establish the value bounds, and then recover the inequality on the original sphere.

3.1 Normalizing a nonzero coefficient

Scaling the roots by t sends (a, b, c, d) to $(t^2 a, t^3 b, t^4 c, t^5 d)$. Algebraic closedness of $\mathbb{C}$ provides $t \neq 0$ with $t^2 a = 1$ if $a \neq 0$, or with $t^3 b = 1$ if $a = 0$, $b \neq 0$. The scale is accounted for through κ and (18). There are three exhaustive cases: $a \neq 0$; $a = 0$, $b \neq 0$; and $a = b = 0$. The first case is split again according to whether its normalized b is zero.

3.2 The generic branch: two quartics

Assume $a = 1$ and $b \neq 0$. Set

$$B = b^2, \quad k = d/b.$$

The three equations (15), expressed in B, c, k , are combined with the saturation equation

$$qB\delta = 1. \quad (19)$$

Since $b \neq 0$ and $\delta \neq 0$, we can take $q = (B\delta)^{-1}$. Polynomial elimination then gives the following conditions.

Lemma 3.1 (Generic elimination). *For coefficient data with $a = 1$ and $b \neq 0$,*

$$P_1(B)P_2(B) = 0, \quad (20)$$

where

$$\begin{aligned} P_1(B) &= 125B^4 + 7900B^3 - 37040B^2 + 44608B + 512, \\ P_2(B) &= 2500B^4 - 114000B^3 + 338925B^2 - 11000B + 6912. \end{aligned}$$

On each branch $P_j(B) = 0$, explicit rational polynomials $C_j, K_j, \Delta_j, R_j, T_j$ satisfy

$$c = C_j(B), \quad k = K_j(B), \quad \delta = \Delta_j(B), \quad r_3 = R_j(B), \quad r_2 = bT_j(B), \quad (21)$$

$$\delta T_j(B)^2 = R_j(B)^3. \quad (22)$$

Certificate proof. Polynomial ideal-membership identities for (15) and (19) yield (20) and the graph equations (21). Substitution into the resultant coefficients and reduction modulo P_j give (22). The coefficient graph polynomials are listed in Appendix A. $\square$

3.3 Constructing the discriminant value polynomials

The branch equations determine Δ_j and R_j by substituting $c = C_j(B)$ and $k = K_j(B)$ into the discriminant and the resultant coefficient r_3 , then taking remainders modulo P_j . We now use these remainders to recover the scale and the objective, rather than the point coordinates.

In $\mathbb{Q}[B]/(P_j)$, compute A_j by reducing $R_j(16\Delta_j)^{-1}$, and compute V_j by reducing $A_j^5\Delta_j$. Both remainders have degree at most three, and polynomial division verifies

$$16\Delta_j A_j - R_j \equiv 0 \pmod{P_j}, \quad A_j^5 \Delta_j - V_j \equiv 0 \pmod{P_j}.$$

Thus, on the corresponding coefficient branch,

$$16\delta A_j(B) = R_j(B), \quad A_j(B)^5 \delta = V_j(B). \quad (23)$$

The explicit cubic polynomials are given in Appendix A. Combining these identities with conjugacy and the discriminant scale relation gives the following value representation.

Lemma 3.2 (Real parameter and discriminant value). *On either generic branch, $B \in \mathbb{R}$ and*

$$A_j(B) = \kappa^2, \quad D^2 = V_j(B)^2. \quad (24)$$

Proof. From (21)–(23) we obtain $\delta r_2^2 = b^2 r_3^3$. Substituting (16) with $a = 1$ gives

$$4096\delta^3 \kappa^6 (\bar{b}^2 - b^2) = 0.$$

The scalar factor is nonzero, so $\bar{b}^2 = b^2$, which is exactly the assertion that B is real. Next, $16\delta A_j = r_3 = 16\delta \kappa^2$ gives $A_j = \kappa^2$. The last relation in (23) becomes $\kappa^{10} \delta = V_j(B)$. Taking squared moduli and using (18) proves (24). $\square$

3.4 Bounds at every real root

Lemma 3.3 (Quartic value bounds). *For $j = 1, 2$ and every $B \in \mathbb{R}$ with $P_j(B) = 0$,*

$$-3125 < V_j(B) < 3125. \quad (25)$$

Certificate proof. Cauchy's polynomial root bound first places every real root of both quartics inside $[-1000, 1000]$. A finite binary subdivision of this interval is checked using exact rational interval arithmetic. At every leaf the checker proves one of two statements: the interval evaluation of P_j excludes zero; or the interval evaluation of V_j lies strictly inside $(-3125, 3125)$. In the former case the leaf contains no root. In the latter case every root in the leaf satisfies the desired value bound.

Interval containment and the fact that the two child intervals cover their parent give the result by induction on the subdivision tree. The two certificates have 55 and 67 nodes, respectively. Their value-bound leaves and the inductive argument are given in Appendix B. The remaining leaves exclude zero from the polynomial range, so the subdivision yields the bound on the entire real zero set without a root count. $\square$

Since $D \geq 0$, Lemmas 3.2 and 3.3 imply $D < 3125$. Indeed $D^2 = V_j(B)^2$ and $|V_j(B)| < 3125$. This contradicts (3), and excludes the entire generic branch from the set of global maximizers.

3.5 Exceptional branches

Lemma 3.4 (Exceptional coefficient data). *The following statements hold for the indicated physical coefficient data.*

1. If $a = 1$ and $b = 0$, then $c = 3/20$, $d = 0$, and $D < 3125$.
2. If $a = 0$ and $b = 1$, the coefficient data are inconsistent.
3. If $a = b = 0$, then $c = 0$ or $d = 0$. The subcase $d = 0$ has $D < 3125$.

Certificate and coefficient proof. For the first two cases, explicit saturated elimination identities imply, respectively, $20c - 3 = 0$, $d = 0$, and the contradiction $1 = 0$. The first case uses 25 certificate steps; the second uses 20. With $a = 1$, $b = 0$, $c = 3/20$, $d = 0$, the equation for r_3 yields $\kappa^2 = 25/4$. Substitution into (18) gives

$$D = \frac{84375}{1024} < 3125.$$

For $a = b = 0$, the s_2 equation reduces to $50000cd^3 = 0$, so $c = 0$ or $d = 0$. In the latter case $\delta = 256c^5 \neq 0$. The r_1 equation gives $\kappa^4|c|^2 = 625/256$. Consequently

$$D^2 = 65536 \left(\frac{625}{256} \right)^5, \quad D = \frac{9765625}{4096} < 3125.$$

$\square$

3.6 The remaining branch and the equality condition

If $a = b = 0$ and $c = 0$, then $p(X) = X^5 + d$. Since $\delta \neq 0$, we have $d \neq 0$. Apply this argument to the unscaled root polynomial of the original maximizer. Its five distinct roots have the same nonzero fifth power. Choose one root u ; the ratios of the roots to u run through all fifth roots of unity. Thus the configuration is a rotated and scaled regular pentagon. All five radii are equal, and the original constraint $N(z) = 5$ forces $|u| = 1$, giving (2).

Proof of Theorem 1.1. Choose a global maximizer on the original sphere. It has positive discriminant, is centered, and produces coefficient data. If $a \neq 0$, normalize to $a = 1$. The branches $b \neq 0$ and $b = 0$ are excluded by Lemmas 3.3 and 3.4. If $a = 0$ but $b \neq 0$, normalize

to $b = 1$ and use the inconsistent branch of Lemma 3.4. If $a = b = 0$, that lemma leaves $c = 0$ or $d = 0$, and $d = 0$ again contradicts the lower bound. The remaining case is a unit regular pentagon. Hence every global maximizer has discriminant 3125, proving the upper bound for every point of the sphere.

If an arbitrary configuration on the sphere has discriminant 3125, the upper bound makes it a global maximizer, so the same classification applies. Conversely a configuration of the form (2) has discriminant 3125, since rotation and permutation preserve the objective and the unit pentagon attains that value. This proves both directions of the equality condition. $\square$

4 Conclusion

We have completed a full Lean 4 [10] formalization of the five-point inequality in Theorem 1.1 and both directions of its equality condition, using mathlib [11]. The proof covers the compactness and stationary reductions, the resultant and mixed-moment identities, the necessary coefficient elimination and scale relations, the strict quartic value bounds, and the recovery of the regular pentagon. Its interval evaluation procedure is proved sound in the computational approach to real bounds described by Melquiond [12]. The complete code is provided at <https://github.com/dgjszy/mordell-five>.

A Explicit polynomials for the generic value comparison

The graph polynomials in Lemma 3.1 are

$$\begin{aligned} C_1(B) &= \frac{99}{446} - \frac{3093}{3568}B + \frac{6015}{14272}B^2 + \frac{375}{57088}B^3, \\ K_1(B) &= \frac{109}{223} - \frac{255}{446}B + \frac{1203}{7136}B^2 + \frac{75}{28544}B^3, \\ C_2(B) &= \frac{48432}{147845} - \frac{65461}{88707}B + \frac{13885}{88707}B^2 - \frac{250}{88707}B^3, \\ K_2(B) &= \frac{224852}{739225} - \frac{776941}{1182760}B + \frac{6436}{29569}B^2 - \frac{285}{59138}B^3. \end{aligned}$$

The scale polynomials and value polynomials are

$$\begin{aligned} A_1(B) &= \frac{67977}{3568} - \frac{195295}{14272}B + \frac{129875}{57088}B^2 + \frac{8375}{228352}B^3, \\ A_2(B) &= \frac{80041741}{7569664} - \frac{12757497155}{545015808}B + \frac{234927875}{34063488}B^2 - \frac{20405375}{136253952}B^3, \\ V_1(B) &= \frac{25821909}{32768} - \frac{109276155}{131072}B + \frac{41181075}{524288}B^2 + \frac{2838375}{2097152}B^3, \\ V_2(B) &= \frac{725183548903125}{496085499904} + \frac{111393859432078125}{3968683999232}B \\ &\quad - \frac{2378884787578125}{248042749952}B^2 + \frac{208919281640625}{992170999808}B^3. \end{aligned}$$

Reduction modulo P_j gives $16\Delta_j(B)A_j(B) = R_j(B)$ and $A_j(B)^5\Delta_j(B) = V_j(B)$ at every root of P_j .

B The interval value certificates

Let $I = [\ell, u]$ have rational endpoints, and let $\text{range}_I(f)$ be the interval obtained by evaluating an expression for f with outward containment using exact rational arithmetic. A leaf is

accepted when

$$0 \notin \text{range}_I(P_j) \quad \text{or} \quad \text{range}_I(V_j) \subset (-3125, 3125).$$

A split at a rational number s replaces I by the two child intervals $[\ell, \min(u, s)]$ and $[\max(\ell, s), u]$. Their union covers the parent, and a split node is accepted when both children are accepted.

Soundness follows by induction. A real zero of P_j lying in an accepted leaf cannot satisfy the first alternative, by interval containment, and therefore satisfies the value bound. At a split it lies in at least one child, to which the induction hypothesis applies. The proof establishes strict endpoint inequalities in the second alternative. Together with the Cauchy bound for the entire real root set, this is a finite proof of Lemma 3.3.

The two subdivisions have 28 and 34 leaves. Respectively, 25 and 31 leaves exclude zero from the range of P_j . The remaining three leaves in each subdivision are displayed in Table 1. Exact rational Horner evaluation gives the listed integer enclosures for V_j ; every endpoint lies strictly between -3125 and 3125 .

Branch	Interval I	Enclosure of $\text{range}_I(V_j)$
P_1	$[\frac{-69375}{1024}, \frac{-34625}{512}]$	$[-3077, -2205]$
P_1	$[\frac{-125}{64}, 0]$	$[788, 2716]$
P_1	$[0, \frac{125}{32}]$	$[-2469, 789]$
P_2	$[\frac{6375}{2048}, \frac{1625}{512}]$	$[539, 2398]$
P_2	$[\frac{1625}{512}, \frac{3375}{1024}]$	$[-2951, 929]$
P_2	$[\frac{347375}{8192}, \frac{2779125}{65536}]$	$[1860, 2636]$

Table 1: Value-bound leaves in the two interval certificates. All remaining leaves exclude roots by interval evaluation of P_j .

The same certificate principle applies to any rational polynomials P, V and positive rational threshold L . Given an interval containing all real zeros of P , an accepted subdivision whose leaves either exclude zero from P or enclose V in $(-L, L)$ proves $|V(\beta)| < L$ at every real zero β of P . The present application uses $P = P_j$, $V = V_j$, and $L = 3125$.

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